

Performance Evaluation of ns-3 Real-Time Emulation

A Comparative Analysis of SimReal-Time and EmuTAP Setups

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- ns-3 is a widely used open-source network simulator.
Actively developed under GNU GPL v2.
- Used in diverse research domains:
computer networks, satellite systems, protocol analysis, etc.
- Provides built-in models for:
 - Ethernet, Wi-Fi, LTE, TCP/UDP, Applications
 - Realistic traffic and topology simulation
- Custom scenarios written in C++ or Python.
- Example: Satellite Network Simulator SNS3 [1]

- Most projects do **not modify** the core of ns-3.
- Instead, they add:
 - New protocols (e.g., QUIC [2])
 - Custom applications and devices
- Community-contributed modules are available on the ns-3 App Store.
- Widely used in academia and industry for rapid experimentation.

Discrete-Event Simulation: Strengths and Limitations

- **Pure simulation mode:** All components (apps + topology) are virtual.
- No real-time constraints: fast-forward or slow-motion possible.
- Ideal for:
 - Controlled experiments
 - Scaling complex scenarios over time
- But...
 - Modeling complex apps can be laborious.
 - Realism vs. simplicity trade-off.
 - Not suitable for Hardware-in-the-Loop (HIL).

Modes of Operation in ns-3

- **Discrete Event Simulation (Native):**
 - Applications + network stack are fully simulated
 - No real-time constraints (runs as fast or slow as needed)
- **Real-Time Simulation Modes:**
 - Time synced to host machine's clock
 - Enables Hardware-in-the-Loop (HIL) integration
 - Useful for testing Quality of Experience (QoE)
- **TAP Integration (EmuTAP):**
 - Real applications communicate through virtual TAP devices
 - Network is simulated by ns-3; apps run on real OS

Source: Ottens et al., 2025 (IEEE Access)

SimReal-Time

- Applications + topology simulated in ns-3
- Runs in real-time mode
- No real applications involved
- Single-threaded execution

EmuTAP

- Real Linux-native apps (e.g., C or Python)
- Simulated network only
- Communication via TAP virtual devices
- Scales with multicore hosts

TABLE 1. Possible configurations of ns-3 testbed setups with real-world integration (adapted from [8]).

Application Nodes	Network Topology		
		Real	Simulated
	Real	Your Computer and Network (a)	<i>ns-3 TAP</i> (b)
	Simulated	<i>ns-3 EMU</i> (c)	<i>ns-3 Native Simulation</i> (d)

TABLE 2. Classification of different ns-3 operation modes.

	Application Code	Network	Time	Reference Table 1
Reality	real	real	real	(a)
ns-3 DCE¹	real	simulated	simulated	-
Sim_{Real-Time}	simulated	simulated	real	(d), but in real-time
Emu_{TAP} (ns-3 TAP)	real	simulated	real	(b)
ns-3 EMU	simulated	real	real	(c)
ns-3 Native	simulated	simulated	simulated	(d)

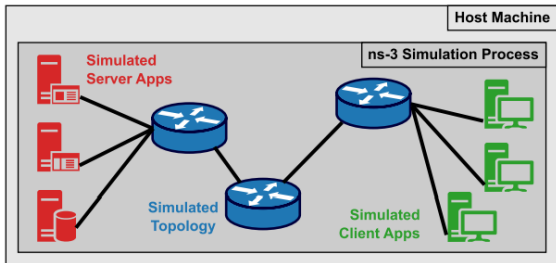


FIGURE 1. Schematic overview of the Sim_{Real-Time} setup for the performance evaluation. Applications as well as network topology are ns-3 models executed in real-time mode. The applications and the routers use the ns-3 protocol stack.

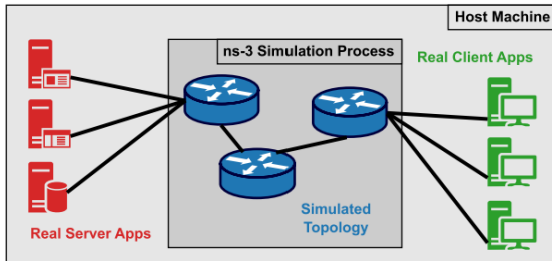


FIGURE 2. Schematic overview of the Emu_{TAP} setup for the performance evaluation. Applications are Linux-native using the Linux protocol stack. The network topology and routers use the ns-3 protocol stack. The interface between Linux-native applications and ns-3 network topology is realized with TAP devices.

Questions?

- [1] P. et al., *Satellite network simulator sns3*, Workshop on Simulation for European Space Programmes (SESP), 2015.
- [2] A. De Biasio *et al.*, “A quic implementation for ns-3,” in *Workshop on ns-3*, 2019.